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Paleomagnetism and the Origin of Mississippi Valley-Type Ore Deposits

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Paleomagnetism can provide useful information about the stratigraphic relationships between the host rocks and the ore of some ore deposits.

Four North American mines with stratabound ore deposits of Mississippi Valley type were sampled and the direction and intensity of the natural remanent magnetization (NRM) were measured. Two of the sites sampled (Newfoundland Zinc Co. property near Daniel's Harbour in western Newfoundland and the St. Joe Minerals Co., #8 Mine in southeast Missouri) had a weak, but measurable NRM in both host and ore rocks. This magnetization proved to be highly stable upon alternating field (AF) demagnetization. The other two mines (Magmont Mine, southeast Missouri, and Pine Point Mine, Northwest Territories, Canada) had intensities of magnetization too low to be measured after demagnetization.

The pole positions computed for the ores and their corresponding hosts are identical within the statistical uncertainty, strongly suggesting that the ore and the host are, geologically speaking, of roughly the same age. This study gives two reliable pole positions, one for late lower Ordovician dolostone and sphalerite ore from Newfoundland of 26° N, 126° E, and the other for the upper Cambrian, based on the Bonneterre dolostone and galena ore from southeast Missouri of 35° S, 170° W.

Within the present limitations of the method the results agree with published opinions concerning the age of the ore, *i.e.* that host rock and ore formation were relatively close in time. Therefore, when significant time differences occur between epigenetic ores and their host rocks, the method may be expected to define this. The method will become progressively more valuable as the apparent polar wandering curves for various continental areas become better defined.

Le paléomagnétisme peut fournir des informations utiles sur les relations stratigraphiques entre les formations rocheuses et le minerai de certains gisements.

On a échantillonné quatre gisements nord-américains du type stratifié de la vallée du Mississipi et on a mesuré la direction et l'intensité du magnétisme naturel remanent (NRM). Deux des endroits échantillonnés (la mine Newfoundland Zinc Co. près de Daniel's Harbour à Terre-Neuve et la mine no. 8 de la compagnie St. Joe Minerals dans le sud-est du Missouri) ont eu un NRM faible mais mesurable dans la roche encaissante et le minerai. Cette aimantation est demeurée très stable malgré une alternance de désaimantation. Les deux autres gisements (Magmont Mine, dans le sud-est du Missouri, et Pine Point Mine, dans les territoires du Nord-ouest, Canada) ont eu des intensités du NRM trop faibles pour pouvoir les mesurer après désaimantation.

Les positions du pôle aussi bien pour les minerais que pour les roches encaissantes sont les mêmes à l'intérieur des limites statistiques ce qui laisse croire que le minerai et la roche encaissante sont géologi-

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quement à peu près du même âge. Cette étude donne deux pôles sur lesquels on peut compter, un pour les calcaires dolomitiques de la fin de l'Ordovicien inférieur et le minerai de zinc de Terre-Neuve qui est de 26° N et 126° E l'autre pour les calcaires dolomitiques de Bonneterre du Cambrien supérieur et le minerai de plomb du sud-est du Missouri qui est de 35° S et 170° W.

A l'intérieur des limites de la méthode, les résultats concordent avec les idées publiées sur l'âge du minerai, *i.e.* que la roche encaissante et le minerai sont à peu près du même âge. En conséquence, lorsqu'on trouve des différences d'âge marquées entre les minerais et la roche encaissante, cette méthode devrait l'indiquer. Elle deviendra de plus en plus valable à mesure que les courbes de migration polaire pour les divers continents seront mieux connues. [Traduit par le journal]

Introduction

Mississippi Valley-type ore deposits commonly occur in relatively undisturbed host sediments that rest upon the continental basement and are therefore ideally suited to paleomagnetic studies. In many cases it is possible to determine that the ore is younger than the enclosing sediments, but it is not possible by field and petrographic methods to determine how much younger. The common occurrence of anomalous leads means that radiometric age estimates based on galena may be misleading (*e.g.*, Ohle 1959; Snyder and Gerdemann 1968; Brown 1970). However, paleomagnetic measurements appear to offer an alternative, independent check based upon apparent polar wandering curves.

The investigations reported here are part of a continuing program that is based upon the concept that paleomagnetic techniques may help to determine the sequence of events involved in the formation of Mississippi Valley-type ore bodies. An earlier preliminary report (Beales and Onasick 1970) indicated that measurable magnetic remanence of very low intensity was present in some host and ore rocks. It has now been shown that some host rocks yield pole positions that are consistent with the known geological time interval. It is possible that in the future we will find some ore bodies in which the ore zone has a magnetic remanence direction which is different from that of the enclosing host rock; in that case the ore would be epigenetic or at least it must have been redistributed or altered some time after the host was deposited.

When, as in the two usable examples from the present study, both host and ore have the same direction of magnetization then either: (*1*) The ore must have been formed at the same time as the host or within a time span sufficiently small as to be unresolvable by paleomagnetic methods; or (*2*) Both host and ore must have been remagnetized together at a later time, possibly at the time of ore emplacement.

The host rocks investigated here have been the subject of many studies and their ages are well established, primarily by the fossil record. The pole position of the host rock could, therefore, be tested to see if it corresponded to the pole expected for rocks of that age. If the pole position is comparable, then the likelihood of secondary magnetization is reduced. If it also can be proved that the ore has a stable magnetization then an age can be assigned to the ore.

In order to test these assumptions, four mines were sampled. These mines are located in Paleozoic platform deposits overlying the Precambrian basement in North America (Fig. 1). Host rock ages range from Cambrian to Devonian. The intensities of magnetization encountered in rocks from the four mines were, in general, very low (see Table 1) and the host rocks measured were consistently higher in intensity of magnetization than the corresponding ores. Two of the mines, Pine Point, Northwest Territories, and Magmont, southeast Missouri, have magnetizations with typical values of $2–4 \times 10^{-8}$ emu cm^{-3}, which become too weak to be measured ($<10^{-8}$ emu cm^{-3}) on demagnetization and no meaningful results could be obtained. Since stability tests are essential, these two mines failed to add definitive data for the study. Thus, conclusions reached here are based on samples collected from the other two mines, Newfoundland Zinc Co. property, Newfoundland, and the St. Joe Minerals Co., #8 Mine, southeast Missouri. These have low, but measurable, remanent magnetization that was stable in alternating fields.

Geological Setting

The mines sampled range from Upper Cambrian (southeast Missouri mines), through Lower Ordovician (Newfoundland), to Middle

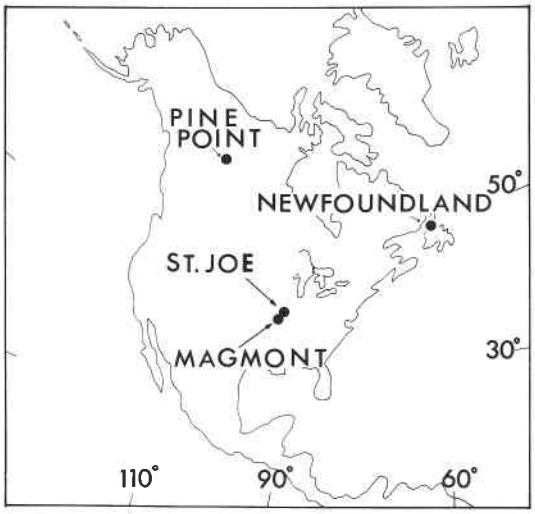


FIG. 1. Paleomagnetic sampling localities for this study.

Devonian (Pine Point, Northwest Territories) in age, (Table 1).

Two mines in the southeast Missouri lead district, located 70 mi (113 km) south of St. Louis, were sampled. The St. Joe Minerals Company, #8 Mine, is in the Old Lead Belt while the Cominco American Co., Magmont Mine, is from the Viburnum (or 'New') Lead Belt, 30 mi (48 km) to the west. The ore deposits are stratiform and samples were collected from flat-lying strata. They are believed to be epigenetic and the Mississippi Valley-type mineralization is believed to be associated with late-lower Ordovician faulting (Snyder and Gerdemann 1968). This means that the geological age difference between host and ore would be small. Paleomagnetic investigations have been carried out on the Lamotte Formation (Al-Khafaji and Vincenz 1971), although only host rocks were sampled by these investigators. The present study is the first to investigate the ores and to comprise both host and ore rocks in these localities.

The ore deposits of southeast Missouri occur within the Upper Cambrian Bonneterre host rock (*e.g.*, Gerdemann and Myers 1972 and references therein). The mineralization occurs in a typical basin marginal facies belt comprising dolomitized algal buildups, oolitic and skeletal calcarenites, backed by more restricted evaporitic facies. This carbonate rock complex forms a fringe around the Precambrian St. Francois Highlands with ore concentrations highest where multiple faulting and fracturing may have connected up diagenetically-modified, primary porosity of the fringing algal reefs and sand bodies with the underlying Lamotte Sandstone aquifer. The host rock consists of a finely sucrosic dolostone in which both traces of original texture and diagenetic features are well preserved. From these observations and the irregular patterns of ore deposition, it is safe to say that some time elapsed between the time of formation of the host rock and the mineralization, but this is believed to be small geologically speaking. If the paleomagnetic results are taken at face value, they provide an additional restraint on theories of ore genesis since it is clear that the ore, while epigenetic, was emplaced not long after host rock genesis. The ore samples studied consisted of massive galena with minor associated pyrite. The remanent magnetism is nonuniformly distributed in the sample although the actual magnetic carriers, presumably minute entrapped and co-precipitated magnetic impurities, have not been identified.

The Newfoundland Zinc Co. ore bodies near Daniel's Harbour, Newfoundland, comprise a series of inter-crystalline and vug-filling sphalerite ore bodies in a *pseudobreccia* dolostone host rock belonging to the St. George Group of Lower Ordovician age (Cummings 1968; Collins and Smith 1972). The ore is clearly epigenetic. It has been genetically related to a pre-middle Ordovician erosion surface by Collins and Smith. If these Newfoundland ores have close affinities with the eastern Tennessee sphalerite deposits in the Lower Ordovician Kingsport/Mascot Formations, and they appear to be similar in many ways, there is some further geological support for a later Lower to Middle Ordovician age for ore deposition (*e.g.*, Maher 1971; Carpenter *et al.* 1971; Hoagland 1971). The paleomagnetic studies reported here tend to support these estimates of timing since it will be noted that the pole positions for both ore and host rock are essentially the same; indicating no great time lapse between deposition of the host limestone, 'preparation of the ground', probably largely by solution due to percolating ground waters, and deposition of the ore from whatever source. No new light is thrown on the mechanism of ore deposition by the paleomagnetic

TABLE 1. Location, petrology, and magnetic remanence intensities of samples

Site	Locality	Geographic coordinates		Age of host rock	Mean magnetic remanence (NRM) ($\times 10^{-8}$ emu cm^{-3})
		Lat.	Long.		
St. Joe #8	Southeast Missouri (U.S.A.)	38° N,	90° W	host: Upper Cambrian—Bonneterre Dolostone	13.7
				ore: Galena	4.4
Newfoundland Zinc	Newfoundland (Canada)	50° N,	57° W	host: Lower Ordovician, St. George Group (dolostone)	87.6
				ore: Sphalerite	17.8
Pine Point Mine	Northwest Territories (Canada)	61° N,	114° W	host: Middle Devonian Pine Point—Presquile Dolostone	3.8
				ore: Sphalerite, galena	2.3
Magmont Mine	Southeast Missouri (U.S.A.)	37° N,	91° W	host: Upper Cambrian—Bonneterre Dolostone	4.5
				ore: Galena	4.0

TABLE 2. Magnetic properties of host rocks and ores

Site	Paramagnetic susceptibility $\chi_p \times 10^{-6}$ (gauss Oe^{-1} g^{-1})	Saturation magnetization J_s (emu g^{-1})	Equivalent Wt % Fe_3O_4*
St. Joe Host	5.60	0.0035	0.0038
St. Joe Ore	−0.77	0.0024	0.0026
Magmont Host	0.30	0.0033	0.0036
Magmont Ore	2.60	0.0017	0.0018
Newfoundland Host	−0.18	0.0028	0.0030
Newfoundland Ore	−0.61	0.0032	0.0034
Pine Point Host	−0.06	0.00021	0.00023
Pine Point Ore	0.17	0.00034	0.00036

*Saturation magnetization at room temperature 92 emu g^{-1}. All ferromagnetism is assumed due to magnetite.

studies. Neither host rock nor ore contain visible magnetic impurities.

Experimental Methods

Magnetic Properties

A set of hysteresis loop measurements were made on representative host and ore samples in order to identify the amount of ferromagnetism as shown in Table 2. Although no direct identification of the ferromagnetic mineral has been made, we have estimated the equivalent amount of magnetite that would be required to be present to account for the ferromagnetism. The amount present is typically less than 0.004% magnetite, an amount comparable to that detected in the Green River Shale (Strangway and McMahon 1973). The high-field susceptibility is also given in Table 2. In some cases the value is positive indicating paramagnetism and is probably due to the presence of Fe^{++} in some of the phases present. In others, the value is negative indicating diamagnetism such as is typical for many natural minerals. Pure galena (PbS), for example, is diamagnetic. The presence of ferromagnetism is therefore due to a small quantity of impurities and the presence of paramagnetism is also due to the presence of small amounts of impurity in the limestones or the ores.

Remanent Magnetization Studies

In general, six oriented hand specimens were collected from host rock and six from ore rocks. Care was taken to select massive, void-filling, ore samples so that essentially no host material would be present. This method of sampling was carried out with a view to doing two-level statistical analysis of the directions of magnetization so that four to six cores were drilled from each oriented specimen.

The direction and intensity of the NRM of the cores were measured using a PAR SM-2D spinner magnetometer. However, due to the very low intensities expected, many precautions were necessary, *e.g.*, averaging of six spins was standard procedure, since strong anisotropies were present, presumably due to the nonuniform distribution of the magnetic materials. The samples were spun inside a magnetically shielded chamber (two layers of molypermalloy) in which the geomagnetic field has been reduced to less than 50 gammas and the background effects of sample cups, holders, etc. subtracted off (see Strangway and McMahon 1973).

In order to minimize the unstable components of the magnetization, AF cleaning was carried out on all the samples, applying alternating fields of 50, 100, 150, 200, and 300 Oe.

Mean directions of magnetizations were computed for each site using the minimum scatter criterion (Irving *et al.* 1961). The best grouping usually occurred at 100 Oe peak alternating field strength. Paleomagnetic directions for each locality were computed for the cleaned mean direction using a two-level Fisher's statistical analysis (Fisher 1953).

The between site precision parameter (K_b) was consistently several times higher than the corresponding within site precision parameter (K_w) determined using both host and ore samples. This suggests statistically that the magnetization of host and ore is in the same direction. This implies that, for the two mines measured, the magnetization of host and ore

was acquired at roughly the same time, within the limitations of the method. Depending on sample site positions relative to the mean continental polar wandering path, it is likely that direction differences could presently be detected only if the age difference between host and ore exceeds about 25 m.y. If it is assumed that the apparent geomagnetic pole has moved from a position near the present equator in Cambrian time to its present position in northern Canada, the apparent wandering has a rate of about 0.1 – 0.2° per million years (Hospers and Van Andel 1967). If direction differences of about 5° are detectable, age differences of about 25 m.y. can be resolved. As instrumentation improves and better apparent polar wandering curves are established, it is possible that somewhat better discrimination will be achieved.

Results

The intensity and stability of the natural remanent magnetization (NRM) of host and ore rocks are summarized in Figs. 3, 4, 5, and 6. In Fig. 2 the intensity of NRM of Newfoundland host and ore are plotted as histograms showing a mean NRM intensity of about 1×10^{-6} emu cm^{-3} for the host and about 2×10^{-7} emu cm^{-3} for the ore. The narrow range of NRM intensity for the host rock (between 1 and 1.5×10^{-6} emu cm^{-3}) differs considerably from the wide distribution of NRM intensities in the ore. In Fig. 3*a* histograms of intensity of NRM are given for six pilot specimens of Newfoundland host and ore rocks for increasing AF peak strengths. These show that both host and ore are highly stable in alternating fields. In Fig. 4*a*, average demagnetization curves of all samples for Newfoundland host and ore are plotted showing that the ore is somewhat more stable to AF demagnetization than the host rock. This seems to be consistently the case in all the examples investigated here. In Fig. 3*b* and 4*b* the corresponding data for the St. Joe Minerals host and ore rocks are given. In this case (Fig. 4*b*) the ore shows a typical initial rise followed by a steady drop in intensity of NRM with increasing fields. This is undoubtedly due to the removal of a super-imposed soft component of opposite sign residing in relatively low coercive domains. Figures 5*a* and 5*b* show the directions of magnetization on a stereonet for all samples before and after AF cleaning. The Newfoundland host (Fig.

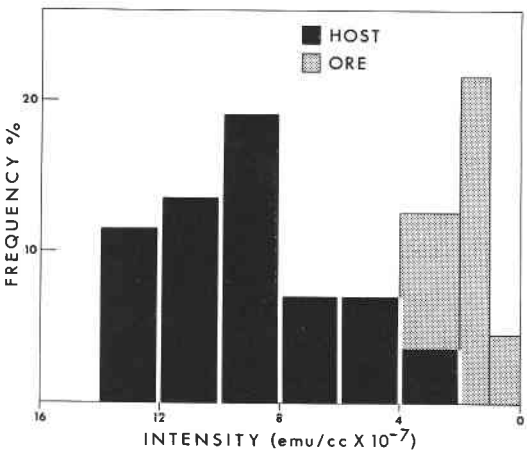


FIG. 2. Histograms of distribution of NRM intensities for Newfoundland host and ore samples (80 samples).

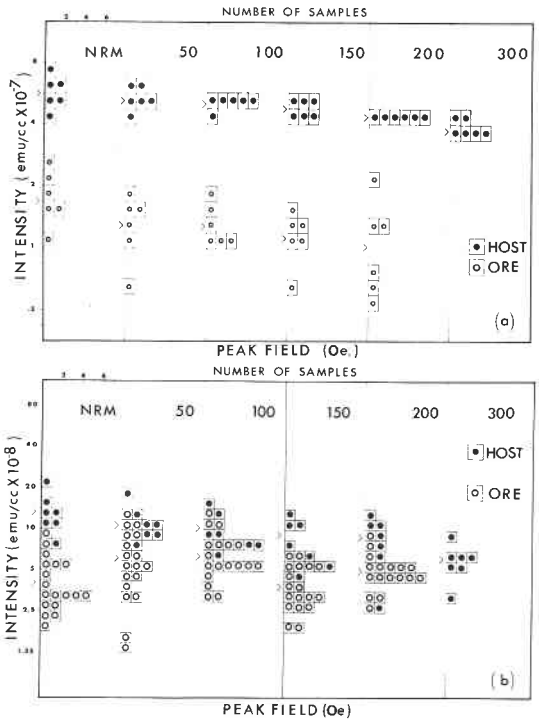


FIG. 3. (*a*) Histograms of distribution of NRM intensities after AF cleaning at peak field strengths of NRM, 50, 100, 150, 200, and 300 Oe for Newfoundland host and ore pilot specimens. Arrows show mean intensities. (*b*) Idem for St. Joe #8 samples.

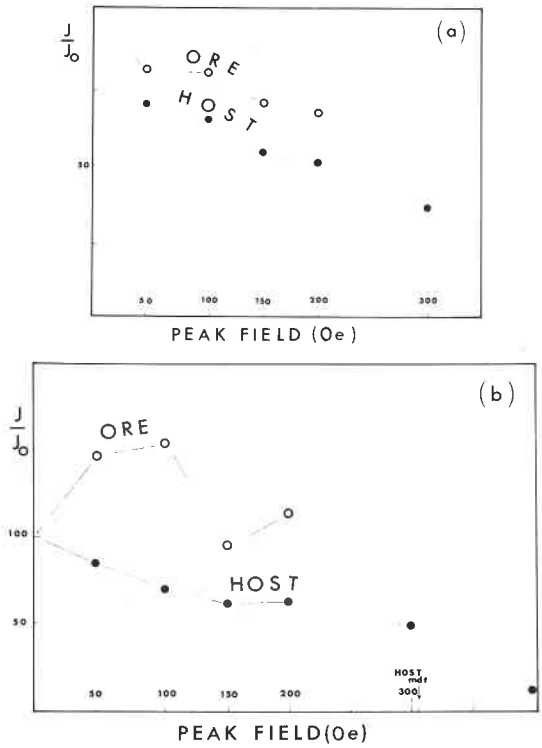


FIG. 4. (*a*) Mean demagnetization curves of Newfoundland samples. (*b*) Idem for St. Joe #8 samples Mean destructive field (mdf) refers to the field required to reduce the initial intensity (J_o) by half.

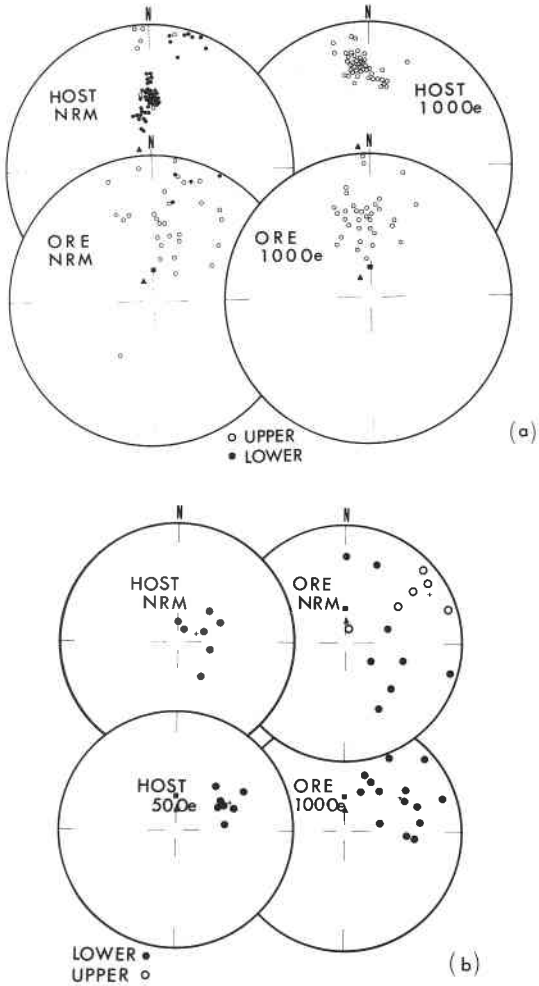


FIG. 5. (*a*) Directions of remanent magnetization of Newfoundland host and ore samples before and after AF demagnetization. Open circles are on the upper hemisphere and solid circles on the lower hemisphere. Stereographic projection. Closed triangles show position of present local field, closed squares that of main dipole. (*b*) Idem for St. Joe #8 samples.

5*a*) shows a fairly good grouping of directions of magnetization before AF cleaning with a mean in a direction intermediate between the present North and the direction of magnetization recovered after cleaning. This is a classic example of the 'smearing' of the directions of magnetization due to a viscous component acquired in the present field, but is easily removed in fields less than 100 Oe. The ore samples are similar in direction before and after cleaning and so do not have a recent viscous component added. This is illustrated in Fig. 6*a*. After AF cleaning at peak field of 100 Oe, all host and ore directions are reversed, the grouping has clearly improved, and both host and ore mean direction of magnetization are close together.

Samples from the St. Joe Minerals #8 Mine show similar results (Fig. 5*b*). The Bonneterre dolostone samples have a downward-pointing magnetization before and after demagnetization, while the ore has both upward and downward direction before AF cleaning, ending with a downward-pointing direction and a mean direction very close to that of the host after demagnetization. In this case the ore has a soft component which is readily removed on AF cleaning. This component is reversed in sense and therefore was not acquired in the present field but at some time in the past when the field was reversed. The changes in mean direction of magnetization for both host and ore are indicated in Fig. 6*b*.

TABLE 3. Paleomagnetic data: Newfoundland (after AF cleaning, at 100 Oe).

	No. of sites	No. of samples	D	I	k_w	k_b	a_{95}	$J \times 10^{-6}$ emu cm^{-3}	M_{100}/M_0	Long.	Lat.	δ_p	δ_m
Host	8	46	358°	−21°	254.6	79.6	5.5°	0.56	0.64	125° E	29° N	3.0°	5.7°
Ore	6	34	357°	−29°	24.5	18.1	13.4°	0.15	0.87	126° E	24° N	8.2°	14.7°
Newfoundland (host and ore)		80	357°	−25°	139.0	202.6	6.9°	—	—	126° E	26° N	4.0°	7.3°

K_w within site precision parameter (mean).
K_b between-site precision parameter.
M_{100}/M_0 ratio of mean intensity, after 100 Oe cleaning, to initial intensity.

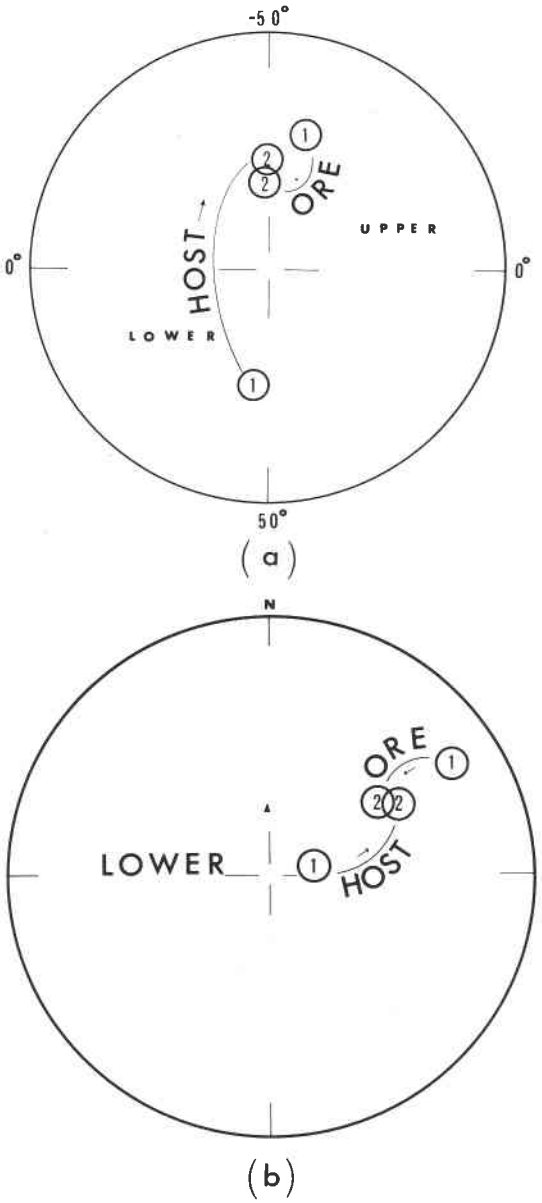


FIG. 6. (*a*) Changes in direction upon AF demagnetization for Newfoundland samples. Mean values. Portion of stereographic projection (equatorial section) 1-NRM, 2-100 Oe. (*b*) Idem for St. Joe #8 samples. Stereographic projection. Triangle shows position of present local field.

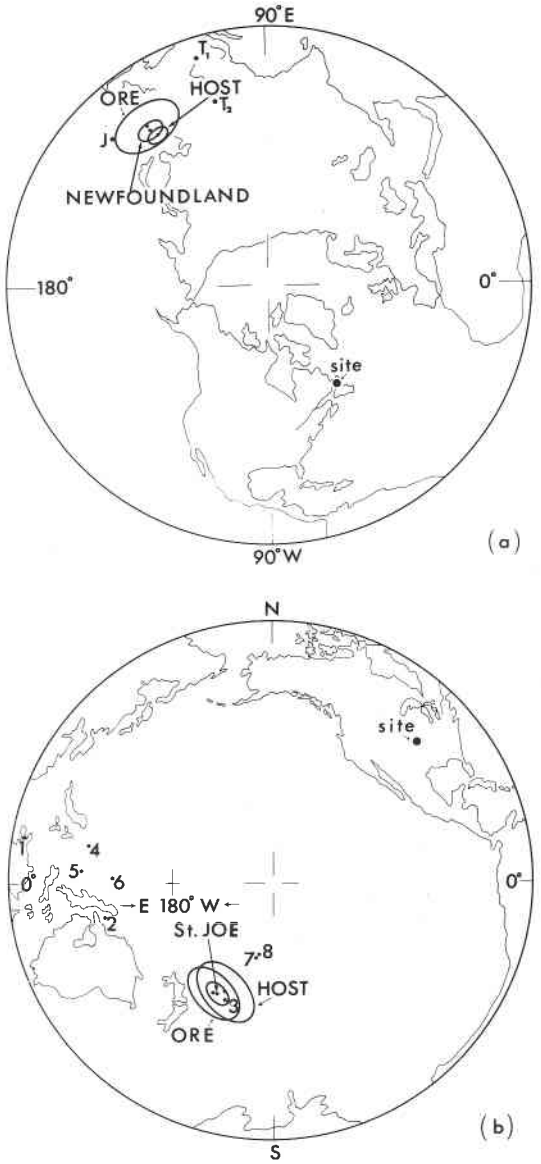


FIG. 7. (*a*) Location of Ordovician pole positions obtained from Newfoundland in present study with other North American Paleozoic data. The pole and 95% confidence ovals for the host, the ore, and Newfoundland site (host and ore combined) are compared with pole positions obtained from published data of North American Ordovician rocks. All polarities are reversed and present data are after AF cleaning. Letters refer to references listed in Table 4. Projection is Lambert's equal area. (*b*) Cambrian pole positions obtained from St. Joe #8 samples, southeast Missouri, compared with published Cambrian data for North America. All St. Joe polarities are normal, and data is after AF demagnetization. Numbers refer to references listed in Table 6. Lambert's equal area projection.

The pole positions computed from the cleaned mean direction of magnetizations along with the ovals of 95% confidence and published pole positions from contemporaneous North American rocks are shown in Fig. 7*a* and Tables 3 and 4 for Newfoundland, and Fig.

TABLE 4. Ordovician pole positions from North America

		Location[1]	Pole position Long.	Lat.	Polarity	Reference
Juniata Formation	Upper Ordovician	J	135° E	20° N	R	Collinson and Runcorn (1960)
Trenton 1	Middle Ordovician	T_1	106° E	9° N	R	Graham (1956)
Trenton 2	Middle Ordovician	T_2	105° E	27° N	R	Graham (1954)
Newfoundland	Middle Ordovician		126° E	26° N	R*	Present study.

*Newfoundland present data is after AF cleaning.
[1]refers to No. of location in Fig. 7*a*.

7*b* and Tables 5 and 6 for southeast Missouri. In the case of southeast Missouri the magnetization of host and ore are virtually indistinguishable. Since both host and ore pole positions fit roughly the polar wandering path for North America in a Cambrian position, a secondary magnetization seems unlikely (see Table 6). The oval of 95% confidence for Newfoundland ore is considerably bigger than the corresponding host oval, but this is probably due to the lower intensity of magnetization of the ore with respect to the host resulting in greater measuring difficulty. On the assumption that host and ore were magnetized when they were formed, and independently, the ore seems to be the same age as the host (within about 25 m.y.). The agreement of this pole position with the data from continental North America implies that any relative rotations between North America and Newfoundland since the Ordovician are small. This implication is of course based on very limited data and much more work needs to be done to define the apparent polar wandering paths.

At the St. Joe Minerals mine (Fig. 7*b* and Table 6, both host and ore pole positions agree roughly with the Cambrian pole position for North America. They are in close agreement with the pole positions indicated for the same locality by Al-Khafaji and Vincenz (1971).

Discussion

This study must be regarded as a preliminary report and while further research is planned we can draw limited conclusions. Apparent polar wandering curves for specific portions of the continental platforms are not known with sufficient accuracy and the carriers of the magnetic remanence in both hosts and ores are presently unknown. Unfortunately the intensities for Pine Point and Magmont Mines ores are too low to permit us to make reliable measurements after demagnetization. However on the positive side, more sensitive magnetometers are being developed and the Newfoundland and Missouri host rocks are sufficiently representative of many deposits, that it is clear that the method will be of general use for dating rocks. The availability of an independent check on some examples of ore-emplacement timing based upon lead isotope abundances is potentially most interesting, since coincident lead and paleomagnetic ages coupled with stratigraphic information will greatly strengthen any conclusions.

The results presented could also be explained if the host rock were magnetized at the time of emplacement of the ore so that the data are not independent samples of the ancient field. Since the intensity of the magnetic remanence in the host rock is an order of magnitude higher than in the ore for both the Newfoundland and the St. Joe Minerals sites and since care was taken in the field to collect host rock samples as remote from ore as possible, a single event is considered less likely than the conclusion that two separate events are involved. However, as noted above, the present results are regarded as preliminary and future sampling procedures for the host rock will be extended to correlative rocks that are even more displaced from the mineralized areas.

Conclusions

(*1*) Some Mississippi Valley-type host rocks and ores carry a natural remanent magnetism (NRM) of low, but measurable intensity. In such cases and when other geological evidence fails to reveal the relative ages of the ore and the host rock, paleomagnetism may provide

TABLE 5. Paleomagnetic data: St. Joe, #8 mine, Southeast Missouri (after AF cleaning, at 100 Oe)

	No. of sites	No. of samples	Remanence							Paleomagnetic pole			
			D	I	k_w	k_b	a_{95}	$J \times 10^{-7}$ emu cm^{-3}	M_{100}/M_0	Long.	Lat.	δ_p	δ_m
Host	2	7	56°	27°	8.2	42.6	15.2°	0.98	.71	167° W	35° S	9.1°	16.2°
Ore	3	14	58°	32°	8.1	59.2	10.5°	0.68	1.54	171° W	35° S	6.7°	12.6°
St. Joe (host + ore)		21	57°	29°	8.1	468.6	4.5°	—	—	170° W	35° S	2.7°	5.3°

NOTE: Symbols as for Table 4.

TABLE 6. Cambrian pole positions from North America

	Location[1]	Pole position Long.	Lat.	Polarity	Reference
Radcliffe–Brook Formation	1	124° E	10° N	R	Black (1964)
Lamotte Formation (Group 11b)	2	155° E	11° S	N	Al-khafaji and Vincenz (1971)
Lamotte Formation (Group 11a)	3	168° W	38° S	N	Al-Khafaji and Vincenz (1971)
Bradore Formation	4	149° E	9° N	R	Black (1964)
Wichita granites	5	147° E	2° N	M	Symons (1967)
Wilbern Formation	6	158° E	0°	?	Howell and Martinez (1957)
Lodore Formation	7	154° W	23° S	M	Collinson and Runcorn (1960)
Tapeats sandstone	8	153° W	22° S	N	Collinson and Runcorn (1960)
St. Joe (late Cambrian)		170° W	35° S	N[2]	Present study

[1]Refers to No. of location in Fig. 7*b*.
[2]All present study data are after AF cleaning.

definable age limits, albeit crude, within which genetic events must have taken place. As more and more sensitive instrumentation becomes available it is possible that the method can be applied to the study of most ore deposits.

(*2*) Pole positions computed from the mean directions of magnetization show that both the Newfoundland and St. Joe Minerals #8 mine hosts and ores have statistically indistinguishable remanent magnetizations. Since the pole positions derived from these mean directions fit the polar wandering path at roughly the position anticipated for the stratigraphic age assigned to the host, a secondary remagnetization of host and ore is unlikely to have occurred, and therefore, the emplacement of the ore seems to be close in a geological time sense to the deposition of the host. A time span of less than 25 m.y. is likely to encompass both the formation of the host and the ore.

(*3*) Reliable pole positions can be obtained from rocks with very low intensity of magnetization.

(*4*) It is possible to imply, from the pole positions obtained from the Newfoundland rocks, that no apparent significant drift of this island with respect to the American continent has occurred since Ordovician time. However, the scarcity of the Ordovician data for North America makes this assumption tenuous at the present time.

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